PES University – Dept. of CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #26 | Smart Cities, Transport & Logistics

Peer-to-Peer Carpooling & Expense Splitter

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Problem Context: A commuter ridesharing portal that matches drivers and passengers with overlapping daily routes, enforces detour tolerance thresholds, and calculates automated fuel cost-share splits.

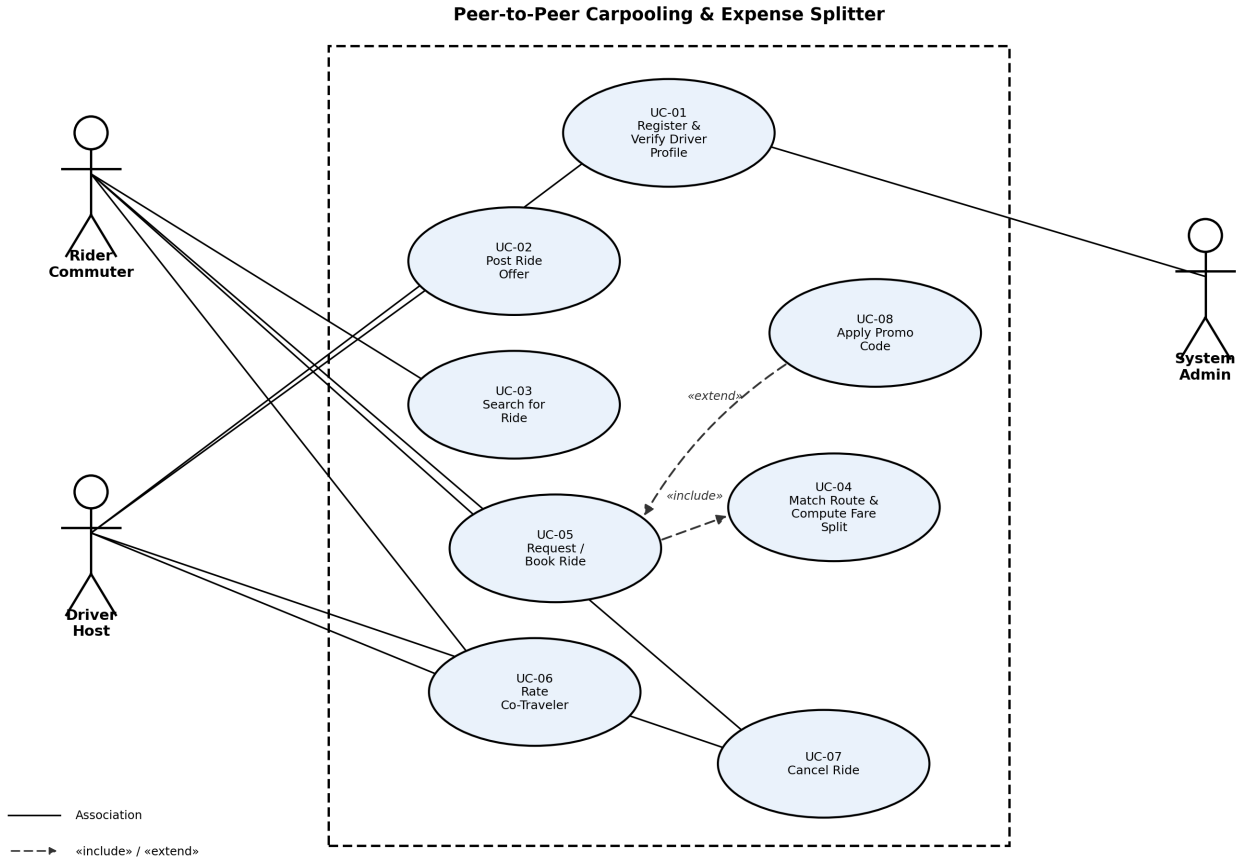
Target Stakeholders / Actors: Rider Commuter, Driver Host, System Admin (license & document verification).

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall calculate route compatibility between a driver's and a passenger's commute paths, permitting matches only within a 1.5 km detour tolerance.	High	Pass: Route match confirmed and fuel cost split calculated when detour ≤ 1.5 km. Fail: A match with detour exceeding 1.5 km is accepted.	Core matching logic; keeps rides efficient for the driver.
FR-002	Functional	The system shall allow a Driver Host to post a ride offer specifying origin, destination, date/time, and number of available seats.	High	Pass: On submission, the ride offer appears in the searchable ride list with all entered fields correct. Fail: Offer is not listed or a field is missing/incorrect.	Enables supply-side inventory of rides for matching.
FR-003	Functional	The system shall allow a Rider Commuter to search for available rides by entering an origin, destination, and preferred travel time window.	High	Pass: Search returns all ride offers within the 1.5 km detour tolerance and time window, ranked by compatibility. Fail: A qualifying ride is omitted or a non-qualifying ride is shown.	Core discovery functionality for riders.
FR-004	Functional	The system shall automatically compute each participant's fuel cost share, proportional to the distance travelled, once a ride is booked.	High	Pass: For a booked ride, the sum of all computed shares equals the total trip fuel cost (± Rs 1 rounding). Fail: Shares do not reconcile with total cost.	Delivers the expense-splitting value proposition.
FR-005	Functional	The system shall allow Rider Commuter and Driver Host to rate and leave feedback for each other after a trip is marked complete.	Medium	Pass: Rating (1–5) is saved and reflected on the counterpart's profile within 5 seconds of submission. Fail: Rating is not saved or not visible on profile.	Builds trust and safety in a peer-to-peer marketplace.
NFR-001	Non-functional (Security)	User ratings, driving-license verification documents, and travel histories must be stored securely with encrypted backups.	High	Pass: Benchmark/security tests confirm AES-256-equivalent encryption at rest and target latency under simulated peak load. Fail: Data found unencrypted or latency target missed.	Protects sensitive personal and identity data; regulatory expectation.
NFR-002	Non-functional (Performance)	The system shall return route-matching search results within 3 seconds for 95% of requests under normal load (up to 500 concurrent users).	Medium	Pass: Load-test logs show 95th-percentile response time ≤ 3 s at 500 concurrent users. Fail: 95th-percentile response time exceeds 3 s.	Ensures a responsive matching experience during peak commute hours.

2. UML Use-Case Diagram

Actors: **Rider Commuter**, **Driver Host**, **System Admin**. The diagram includes an **«include»** relationship (Request/Book Ride always includes Match Route & Compute Fare Split) and an **«extend»** relationship (Apply Promo Code optionally extends Request/Book Ride).



3. Use-Case Flow Specification

Use Case: UC-05 – Request / Book Ride

Primary Actor: Rider Commuter
Secondary Actor: Driver Host, System (fare-split engine)

Preconditions:

- The Rider Commuter is registered and logged into the system.
- At least one Driver Host has an active, published ride offer (UC-02) for the relevant date.

Postconditions:

- A confirmed booking record exists linking the Rider Commuter to the Driver Host's ride.
- The fuel cost share for each participant has been calculated and displayed (via included UC-04).
- The Driver Host's available seat count is decremented by one.

Main Success Scenario:

1. Rider Commuter searches for a ride using origin, destination, and preferred time window (UC-03).
2. System displays a list of candidate ride offers within the 1.5 km detour tolerance.
3. Rider Commuter selects a ride offer and taps “Request to Book”.
4. System *includes* UC-04: it re-validates route compatibility and computes the proportional fuel cost split for all participants.
5. System displays the computed fare share to the Rider Commuter for confirmation.
6. Rider Commuter confirms the booking.
7. System notifies the Driver Host of the new booking request.
8. Driver Host accepts the request.
9. System confirms the booking to both parties, decrements available seats, and marks the use case complete.

Alternate Flow – 4a. Detour Tolerance Exceeded / No Compatible Match:

4a1. During the include-invocation of UC-04, the system determines the selected route exceeds the 1.5 km detour tolerance (e.g. the Driver Host edited their route after the search was performed).
4a2. System displays a “Route no longer compatible” message and cancels the pending request.
4a3. System returns the Rider Commuter to the search results (UC-03), excluding the incompatible offer.
4a4. Rider Commuter selects an alternate ride offer or exits the use case.

Alternate Flow – 8a. Driver Host Declines Request:

8a1. Driver Host declines the booking request within the response window.
8a2. System notifies the Rider Commuter that the request was declined and releases the held seat.
8a3. Rider Commuter is returned to the search results (UC-03) to select another ride.